



UNIVERSITEIT•STELLENBOSCH•UNIVERSITY
jou kennisvennoot • your knowledge partner

Development of a High-Power Audio Amplifier with an Automated Thermal Management System

by

Karl Johan Hofmeyr
27150526

Mechatronic Project 478

Final Report

Study leader: Ms T. Hall

2026

Declaration

I have read and understand the Stellenbosch University Policy on Plagiarism and the definitions of plagiarism and self-plagiarism contained in the Policy.

I also understand that direct translations are plagiarism, unless accompanied by an appropriate acknowledgement of the source. I also know that verbatim copy that has not been explicitly indicated as such, is plagiarism.

I know that plagiarism is a punishable offence and may be referred to the University's Central Disciplinary Committee (CDC) who has the authority to expel me for such an offence.

I know that plagiarism is harmful for the academic environment and that it has a negative impact on any profession.

Accordingly all quotations and contributions from any source whatsoever (including the internet) have been cited fully (acknowledged); further, all verbatim copies have been expressly indicated as such (e.g. through quotation marks) and the sources are cited fully.

I declare that, except where a source has been cited, the work contained in this assignment is my own work and that I have not previously (in its entirety or in part) submitted it for grading in this module/assignment or another module/assignment.

I declare that I have not allowed, and will not allow, anyone to use my work (in paper, graphics, electronic, verbal or any other format) with the intention of passing it off as his/her own work.

I know that a mark of zero may be awarded to assignments with plagiarism and also that no opportunity may be given to submit an improved assignment.

Name: Karl Johan Hofmeyr

Student no: 27150526

Signature: _____

Date: July 22, 2026

MECHATRONIC PROJECT 478: EXECUTIVE SUMMARY

Student: K.J. Hofmeyr

Title of Project
The Development of a High-Power Audio Amplifier with an Automated Thermal Management System
Objectives
Design and Build an Audio Amplifier Capable of Driving an 8Ω Speaker at 75 W_{RMS} for an Extended Period of Time, with Thermal Monitoring and Management
What is current practice and what are its limitations?
What is new in this project?
If the project is successful, how will it make a difference?
What are the risks to the project being a success? Why is it expected to be successful?
What contributions have/will other students made/make?
Which aspects of the project will carry on after completion and why?
What arrangements have been/will be made to expedite continuation?

Student

Date

Lecturer

Abstract

Die projek behels die ontwerp, bou en toets van 'n vibrasie toetsbank. Die invloed wat 'n vibrasie het op die trekkrag wat op 'n ploeg uitgeoefen word moet gesimuleer word. Die toetsbank simuleer so 'n situasie.

Die toetsbank moet ook aanpasbaar wees, sodat ...

Contents

Declaration	i
Mechatronic Project 478: Executive Summary	ii
Abstract	iii
Contents	iv
List of Figures	v
List of Tables	vi
Nomenklatuur	vii
1 Literature Review	1
1.1 Introduction	1
1.2 Audio Signal Fundamentals	1
1.2.1 Audio Frequency Range	1
1.2.2 Distortion and Noise	1
1.3 Audio Equalisation	3
A Konsepte Gegener eer	5
A.1 Konsep I	5
A.2 Konsep II	5
List of References	6

List of Figures

1.3.1	Effect of Q on bandwidth for a graphic equalizer	4
-------	--	---

List of Tables

Nomenklatuur

Konstantes

$$g = 9.81 \text{ m/s}^2$$

Veranderlikes

Re_D	Reynolds getal t.o.v. deursnit	[]
x	Koordinaat	[m]
$\ddot{x}$	Versnelling	[m/s ²]
θ	Rotasiehoek	[rad]
τ	Moment	[N·m]

Vektore en tensore

$\vec{v}$ Fisiese vektor, sien vergelyking ...

Subscripts

a	Adiabaties
a	Koordinaat

Abbreviations and Acronyms

SID Slew-Induced Distortion

Chapter 1

Literature Review

1.1 Introduction

Hierdie projek vloei uit 'n voorstel van Mnr D.N.J. Els en vorm ook deel van ...

1.2 Audio Signal Fundamentals

1.2.1 Audio Frequency Range

The audible frequency bandwidth of the human ear is widely considered to be 20 Hz to 20 kHz, (Self, 2009, p. 20) and (Slone, 1999, p. 42). One would assume that when designing an audio amplifier you would take only this range into consideration, but this is untrue because of a phenomenon known as *slew-induced distortion* or SID. When an amplifier cannot change its output voltage fast enough because of a rapidly changing or high frequency input signal, it tends to cause distortion at lower frequencies. For example, an inaudible 30 kHz signal could distort and interact with amplifier non-linearity, produce audible unwanted byproducts, (Slone, 1999, p. 43). We therefore should design our amplifier with a higher frequency bandwidth in mind.

1.2.2 Distortion and Noise

Linearity describes the extent to which the relationship between a circuit's input and output can be represented by a straight transfer characteristic (Self, 2009, p. 428). An ideal power amplifier's transfer function should be represented by a straight line in the form:

$$v_{out} = A_v v_{in} \tag{1.2.1}$$

Whereas a non-linear amplifier would behave approximately as:

$$v_{\text{out}} = a_1 v_{\text{in}} + a_2 v_{\text{in}}^2 + a_3 v_{\text{in}}^3 + \dots \quad (1.2.2)$$

The higher-order terms cause the gain to vary with instantaneous signal amplitude, thereby changing the waveform shape and generating harmonic or intermodulation distortion.

Distortion is unwanted change that an amplifier makes to the shape of an input signal. There are several sources of distortion in a power audio amplifier. The seven main causes of distortion are because of non-linearity in either the input stage, *Voltage Amplification Stage* or VAS, output stage, non-linear input impedance loading of the VAS, rail decoupling, induction of supply currents or NFB feed taken from the wrong place, (Self, 2009, p. 65-67).

Input Stage Distortion

Input stage distortion can be reduced by adding emitter degeneration, (Self, 2009, p. 79-80), balancing the differential input pair, (Self, 2009, p. 80-82) or increasing collector current whilst maintaining transconductance with emitter degeneration resistors (Self, 2009, p. 86-87).

VAS Distortion

VAS distortion is caused by the curved qualities of the nonlinear $V_{BE}-I_C$ transfer characteristic. The VAS can be linearised by increasing the local open loop gain to obtain greater local *Negative Feedback* or NFB, Self (2009).

Output Stage Distortion

There are three main causes for distortion in the output stage, one of which is *large-signal non-linearity* or LSN. LSN occurs when the amplifier's output devices must handle large voltage or current swings and their electrical behaviour is no longer constant. LSN can be reduced by increasing the number of output devices or choosing higher quality output devices (Self, 2009, p. 164-166).

The second is crossover distortion, which occurs a Class-B push-pull output stage during the transfer of control between the positive and negative output devices. This is eliminated with a Class-AB amplifier by applying a slight bias at your output stage.

The third is switching distortion which occurs when the output transistor that was conducting does not switch off quickly enough as the signal changes polarity. This can be fixed by using an output topology capable of reverse-biasing the output-transistor base-emitter junction during turn-off, particularly the EF Type II or

using lower driver emitter or collector resistance so that stored charge is removed more quickly, (Self, 2009, p. 185).

VAS-Loading Distortion

VAS-Loading Distortion is caused by the non-linear loading of the output stage.

1.3 Audio Equalisation

The main purpose of an audio equaliser circuit is to balance the different frequency bands present in an audio signal. For this project, it is required to design a equaliser centered around at least three frequency bands, the bass-, mid- and treble frequencies.

The graphic equaliser design from the Texas Instrument article, Carter (2001), has been followed. The design uses simulated inductors and the article expands on equaliser qualities, such as the the bandwidth, gain and Q-values of the equaliser.

A simulated inductor reverses the operation of a capacitor using an operational amplifier (Carter, 2001, p. 34). Obtaining real inductors is expensive and difficult whereas a simulated inductor is inexpensive and has a smaller footprint.

Bandwidth is the range of frequencies affected by a one equaliser band and it is inversely correlated to its Q value. A higher Q value therefore displays a narrower, more pointy bandwidth as shown in Figure 1.3.1 from (Carter, 2001, p. 37).

The Q value is determined by the formula:

$$Q = \frac{X_L}{R} \quad (1.3.1)$$

Where the inductive reactance is determine by:

$$X_L = 2\pi \times f_O \times L \quad (1.3.2)$$

And the resonant frequency and simulated inductor is calculated with:

$$f_O = \frac{1}{2\pi\sqrt{LC}} \quad (1.3.3)$$

$$L = (R5 - R4) \times R4 \times C1 \quad (1.3.4)$$

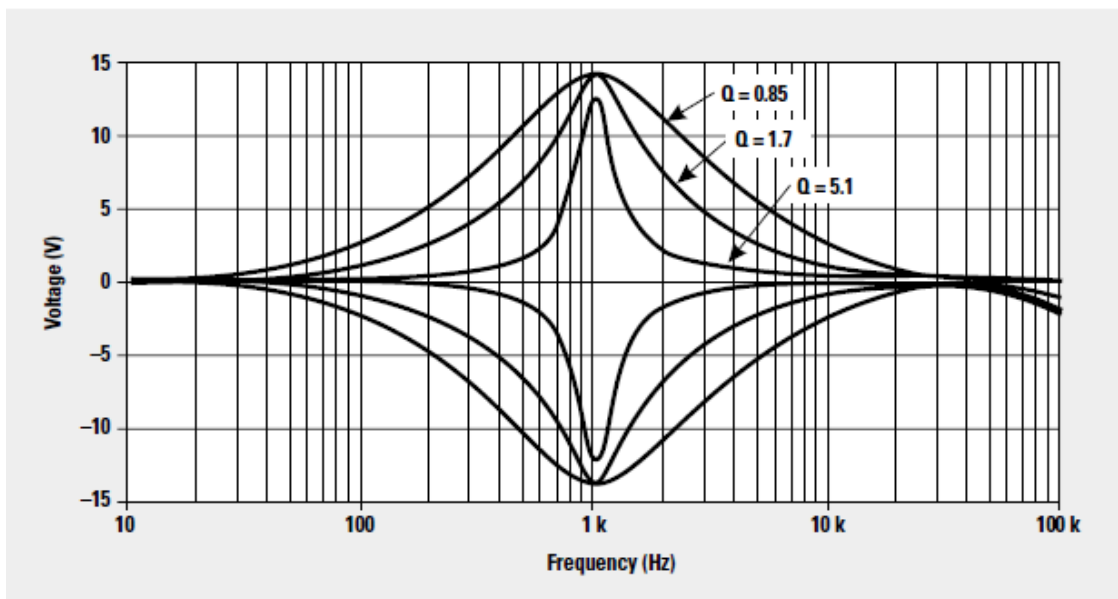


Figure 1.3.1: Effect of Q on bandwidth for a graphic equalizer

Appendix A

Konsepte Gegenereer

A.1 Konsep I

A.2 Konsep II

List of References

- Carter, B. (2001 July). An audio circuit collection, part 3. *Analog Applications Journal*, pp. 34–39.
- Self, D. (2009). *Audio Power Amplifier Design Handbook*. 5th edn. Focal Press. ISBN 978-0-240-52162-6.
- Slone, G.R. (1999). *High-Power Audio Amplifier Construction Manual: 50 to 500 Watts for the Audio Perfectionist*. McGraw-Hill, New York. ISBN 978-0-07-134119-6.